

**Identifying Glaucoma Patients Requiring Further Psychiatric Evaluation Using An
Electronic Health Record–Based Indicator**

Master Project Proposal

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I. Brief Change Since Last Update

Since the submission of the Specific Aims page, the project has been refined to focus on a fully supervised grouped encoder–classifier framework trained on the silver-standard label (r). The r label now serves as a screening indicator to identify glaucoma patients who may require further psychiatric evaluation. The previously proposed semi-supervised and entropy-regularized training components were removed to simplify the modeling framework and strengthen interpretability. In addition, we will now evaluate the trained model using a sequential diagnostic testing design, which will include evaluating standard operating characteristics (e.g., positive predictive value), and will also include SHAP-based feature interpretation.

II. Background and Significance:

Psychiatric distress, including depression and anxiety, is common among patients with vision-threatening eye diseases such as glaucoma. Distress can reduce treatment adherence, impair quality of life, and worsen long-term outcomes. Yet routine psychological screening is rarely performed in ophthalmology settings due to time and staffing limitations.

Recent work by Berchuck et al. (TVST 2022) demonstrated that routinely collected EHR data can be leveraged to identify patients with elevated psychiatric distress risk, achieving strong predictive performance using machine learning methods. Their findings highlighted that both psychiatric and systemic health factors—such as chronic pain, cardiovascular, or metabolic disorders—carry predictive value, emphasizing the interconnected nature of mental and physical health.

Building on this evidence, our project aims to develop a fully supervised grouped encoder–classifier model to determine, from structured EHR data, whether a glaucoma patient may require further psychiatric evaluation. In this framework, the r label represents an algorithmic, silver-standard indicator of possible distress. Our model is not meant to replace clinical assessment but to prioritize patients for follow-up screening, thereby improving mental-health detection efficiency within ophthalmology practice.

III. Learning Objectives

1. Develop proficiency in designing and training modular neural network architectures for structured EHR data, including group-specific encoders and shared classifiers.
2. Apply advanced model evaluation and interpretability tools such as ROC analysis, [latent space visualization](#), and SHAP explanations.

3. Translate machine-learning outputs into clinically meaningful insights, using the r label as a screening indicator to identify glaucoma patients who may need further psychiatric evaluation.
4. Strengthen statistical communication and reporting skills through preparation of a written report, presentation slides, and committee meetings.

IV. *Specific Aims and Method*

- Aim 1 – Build a Baseline Supervised Model for Distress Condition among Glaucoma Patients
 - i. Objective: Establish an initial benchmark for predicting the r distress indicator using routinely collected EHR data.
 - ii. Approach: We trained a Random Forest classifier on the labeled glaucoma cohort to obtain a basic understanding of predictive signal in the data. Hyperparameters were selected through cross-validation, and model performance was evaluated on a patient-level held-out test set using the area under the ROC curve (AUROC) and the confusion matrix.
 - iii. Outcome: The baseline model demonstrated that routinely collected EHR data contain measurable predictive information for identifying patients who may require further psychiatric evaluation. However, the model failed to capture the complex structure and cross-domain interactions inherent in EHR data, motivating the development of a grouped encoder–classifier framework in Aim 2.
- Aim 2 – Develop a Grouped Encoder–Classifier Model to Capture Structured Relationships in EHR Data
 - i. Objective: Address the limitations of traditional models by developing a modular neural network that captures structured relationships across distinct EHR domains for predicting the r distress indicator.
 - ii. Approach: Each data domain—diagnoses, medications, procedures, and demographics—was processed through its own encoder to learn domain-specific latent representations. These representations were concatenated into a unified latent vector, which was then passed to a shared two-layer MLP classifier trained with binary cross-entropy loss on the r label. Model training employed early stopping and validation monitoring to prevent overfitting. The performance is evaluated by sequential post analysis.
 - iii. Outcome: The grouped encoder–classifier achieved higher discrimination and improved calibration compared with the baseline Random Forest model. This trained architecture now serves as the foundation for Aim 3, where a temporal CNN component will be incorporated to extract encounter-level time-window features.
- Aim 3 – Incorporate Temporal Feature Extraction Using Convolutional Neural Networks
 - i. Objective: Extend the grouped encoder–classifier model to capture longitudinal patterns within each patient’s EHR data by integrating a convolutional neural network (CNN) that learns from encounter-level time

windows. This enhancement aims to detect temporal trends—such as evolving diagnostic or medication histories—that may precede psychiatric distress.

- ii. **Approach:** We will integrate a convolutional neural network (CNN) component to summarize encounter-level EHR data across defined time windows. This will allow the model to capture changes in diagnoses, medications, or procedures that may indicate increasing psychiatric risk. The CNN will operate on longitudinal representations produced by the grouped encoders and feed its summarized output into the shared classifier.
- iii. **Outcome:** Incorporating temporal information is expected to improve prediction accuracy and provide a more dynamic understanding of how patient health trajectories relate to distress risk, supporting earlier and more targeted psychiatric evaluation.

V. Timeline

- **Aim 1 – Baseline Model Development**
 - i. **Month 1:** Begin data preprocessing and train baseline Random Forest model to evaluate predictive signal for the r indicator.
 - ii. **Month 2:** Complete cross-validation and summarize model performance.
 - iii. **Month 3:** Interpret findings and identify the need for a more structured modeling framework.
- **Aim 2 – Grouped Encoder–Classifier Model**
 - i. **Month 4:** Develop and train grouped encoder–classifier model with domain-specific encoders.
 - ii. **Month 5:** Evaluate performance and confirm improvement compared with the baseline model.
- **Aim 3 – Incorporate Temporal Features Using CNN**
 - i. **Month 6:** Begin integrating CNN component to capture temporal patterns across patient encounters.
 - ii. **Month 7:** Complete CNN model training and conduct evaluation against the grouped encoder–classifier.
 - iii. **Month 8:** Perform interpretability analyses (SHAP, calibration, visualization).
 - iv. **Month 9:** Revise analyses and complete written report and presentation slides.
 - v. **Month 10 (Tentative):** Conduct **Final Defense (MP-4)** and submit final written report.

VI. Tentative Date for Midterm progress report: January 12, 2026

VII. Tentative Date for Final Defense and Presentation: March 16, .2026